LIMESTONE AND KARST LANDFORMS / INDIA CAVES-MEGHALAYAN AGE

KARST COUPLED SYSTEM → CARBONATE CYCLE → KARST PREREQUISITES → HYDROLOGICAL ZONES → SURFACE-KARST LADDER → CAVE-DEVELOPMENT SEQUENCE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

● PRIMARY CORE FLOW ● SUBORDINATE EXTRA ● APPROVAL / REVIEW

Approval: FALSE • Pending user review • source generation g4 and all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles

00

STAGE 00 KARST COUPLED SYSTEM

KARST COUPLED SYSTEM | JOINTS, BEDDING AND SINK POINTS | OPENINGS ENLARGE UNDERGROUND | SPRINGS EXPORT WATER AND SOLUTE | SURFACE + CAVE | ONE HYDROLOGICAL LANDSCAPE | MUST REMEMBER | KARST REQUIRES SOLUBLE ROCK, JOINTS, WATER AND CARBONIC-ACID SOLUTION

RECHARGE → joints, bedding and sink points → DISSOLUTION → openings enlarge underground → CONDUITS → springs export water and solute → SURFACE + CAVE → one hydrological landscape → MUST REMEMBER: Karst requires soluble rock, joints, water and carbonic-acid solution, followed by underground drainage

CORE CLAIM
- RECHARGE → joints, bedding and sink points
- DISSOLUTION → openings enlarge underground

MECHANISM
- CONDUITS → springs export water and solute
- SURFACE + CAVE → one hydrological landscape

CONSEQUENCE / LIMIT
- MUST REMEMBER: Karst requires soluble rock, joints, water and carbonic-acid solution, followed by underground drainage and carbonate deposition
- lapies, sinkholes, dolines, uvalas, poljes, caves, stalactites and stalagmites form a connected hydrological system.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MUST REMEMBER: Karst requires soluble rock, joints, water and carbonic-acid solution, followed by underground drainage and carbonate deposition; lapies, sinkholes, dolines, uvalas, poljes, caves, stalactites and stalagmites form a connected hydrological system.

01

STAGE 01 CARBONATE CYCLE

CARBONATE CYCLE | CO_2 + WATER | WEAK CARBONIC ACID | ACID + $CaCO_3$ | DISSOLVED BICARBONATE | SOLUTION MOVES THROUGH CONDUITS | CALCITE CAN PRECIPITATE

CO_2 + WATER → weak carbonic acid → ACID + $CaCO_3$ → dissolved bicarbonate → TRANSPORT → solution moves through conduits → DEGASSING → calcite can precipitate

CORE CLAIM
- CO_2 + WATER → weak carbonic acid
- ACID + $CaCO_3$ → dissolved bicarbonate

MECHANISM
- TRANSPORT → solution moves through conduits
- DEGASSING → calcite can precipitate

CONSEQUENCE / LIMIT
- CO_2 + WATER → weak carbonic acid
- ACID + $CaCO_3$ → dissolved bicarbonate

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: TRANSPORT → solution moves through conduits; DEGASSING → calcite can precipitate.

02

STAGE 02 KARST PREREQUISITES

KARST PREREQUISITES | SOLUBLE, THICK AND EXPOSED TO RECHARGE | JOINTS AND BEDDING PATHWAYS | GRADIENT PLUS OUTLET | BASE LEVEL | SUSTAINED DISSOLUTION AND REMOVAL

ROCK → soluble, thick and exposed to recharge

STRUCTURE → joints and bedding pathways → FLOW → gradient plus outlet / base level → TIME → sustained dissolution and removal

CORE CLAIM
- ROCK → soluble, thick and exposed to recharge
- STRUCTURE → joints and bedding pathways

MECHANISM
- FLOW → gradient plus outlet / base level
- TIME → sustained dissolution and removal

CONSEQUENCE / LIMIT
- ROCK → soluble, thick and exposed to recharge
- STRUCTURE → joints and bedding pathways

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOW → gradient plus outlet / base level.

03

STAGE 03 HYDROLOGICAL ZONES

HYDROLOGICAL ZONES | SHALLOW STORAGE AND ROUTING | DESCENDING WATER IN AIR-FILLED VOIDS | WATER TABLE | FLUCTUATING BOUNDARY | SATURATED TUBES AND LOOPS

EPKARST → shallow storage and routing

VADOSE → descending water in air-filled voids → WATER TABLE → fluctuating boundary → PHREATIC → saturated tubes and loops

- ROCK → soluble, thick and exposed to recharge
- STRUCTURE → joints and bedding pathways

- FLOW → gradient plus outlet / base level
- TIME → sustained dissolution and removal

- ROCK → soluble, thick and exposed to recharge
- STRUCTURE → joints and bedding pathways

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOW → gradient plus outlet / base level.

03

STAGE 03

HYDROLOGICAL ZONES

HYDROLOGICAL ZONES · SHALLOW STORAGE AND ROUTING · DESCENDING WATER IN AIR-FILLED VOIDS · WATER TABLE · FLUCTUATING BOUNDARY · SATURATED TUBES AND LOOPS

EPKARST → shallow storage and routing

VADOSE → descending water in air-filled voids → WATER TABLE → fluctuating boundary → PHREATIC → saturated tubes and loops

CORE CLAIM
- EPKARST → shallow storage and routing
- VADOSE → descending water in air-filled voids

MECHANISM
- WATER TABLE → fluctuating boundary
- PHREATIC → saturated tubes and loops

CONSEQUENCE / LIMIT
- EPKARST → shallow storage and routing
- VADOSE → descending water in air-filled voids

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: WATER TABLE → fluctuating boundary; PHREATIC → saturated tubes and loops.

04

STAGE 04

SURFACE-KARST LADDER

SURFACE-KARST LADDER · SMALL SOLUTION GROOVES · CLOSED DEPRESSION · LARGER COMPOSITE · STRUCTURAL BASIN · SWALLOW HOLE · SURFACE STREAM TRANSFERS UNDERGROUND

KARREN → small solution grooves

DOLINE → closed depression → UVALA / POLJE → larger composite / structural basin → SWALLOW HOLE → surface stream transfers underground

CORE CLAIM
- KARREN → small solution grooves
- DOLINE → closed depression

MECHANISM
- UVALA / POLJE → larger composite / structural basin
- SWALLOW HOLE → surface stream transfers underground

CONSEQUENCE / LIMIT
- KARREN → small solution grooves
- DOLINE → closed depression

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: UVALA / POLJE → larger composite / structural basin.

05

STAGE 05

CAVE-DEVELOPMENT SEQUENCE

CAVE-DEVELOPMENT SEQUENCE · FAVOURABLE FRACTURE OR BEDDING PLANE · PASSAGE ENLARGES UNDER SATURATION · STREAM INCISES AFTER WATER-TABLE FALL · LATE CHANGE · COLLAPSE, FILL OR RENEWED SOLUTION

INCEPTION → favourable fracture or bedding plane → PHREATIC → passage enlarges under saturation → VADOSE → stream incises after water-table fall → LATE CHANGE → collapse, fill or renewed solution

CORE CLAIM
- INCEPTION → favourable fracture or bedding plane
- PHREATIC → passage enlarges under saturation

MECHANISM
- VADOSE → stream incises after water-table fall
- LATE CHANGE → collapse, fill or renewed solution

CONSEQUENCE / LIMIT
- INCEPTION → favourable fracture or bedding plane
- PHREATIC → passage enlarges under saturation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LATE CHANGE → collapse, fill or renewed solution.

06

STAGE 06

SPELEOTHEM MAP

SPELEOTHEM MAP · ROOF DRIP DEPOSIT · FLOOR SPLASH · DRIP DEPOSIT · ROOF AND FLOOR FORMS JOIN · FILM FLOW ON WALL OR SLOPE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
STALACTITE → roof drip deposit	COLUMN → roof and floor forms join	STALACTITE → roof drip deposit
STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope	STALAGMITE → floor splash / drip deposit

CORE CLAIM
- STALACTITE → roof drip deposit
- STALAGMITE → floor splash / drip deposit

MECHANISM
- COLUMN → roof and floor forms join
- FLOWSTONE / CURTAIN → film flow on wall or slope

CONSEQUENCE / LIMIT
- STALACTITE → roof drip deposit
- STALAGMITE → floor splash / drip deposit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LATE CHANGE → collapse, fill or renewed solution.

06

STAGE 06

SPELEOTHEM MAP

SPELEOTHEM MAP · ROOF DRIP DEPOSIT · FLOOR SPLASH · DRIP DEPOSIT · ROOF AND FLOOR FORMS JOIN · FILM FLOW ON WALL OR SLOPE

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
STALACTITE → roof drip deposit	COLUMN → roof and floor forms join	STALACTITE → roof drip deposit
STALAGMITE → floor splash / drip deposit	FLOWSTONE / CURTAIN → film flow on wall or slope	STALAGMITE → floor splash / drip deposit

CORE CLAIM

- STALACTITE → roof drip deposit
- STALAGMITE → floor splash / drip deposit

MECHANISM

- COLUMN → roof and floor forms join
- FLOWSTONE / CURTAIN → film flow on wall or slope

CONSEQUENCE / LIMIT

- STALACTITE → roof drip deposit
- STALAGMITE → floor splash / drip deposit

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FLOWSTONE / CURTAIN → film flow on wall or slope.

07

STAGE 07

INDIA CAVE FIREWALL

INDIA CAVE FIREWALL · MEGHALAYA LIMESTONE · NATURAL CAVES IN ANDHRA PRADESH · HUMAN ROCK-CUT ARCHITECTURE · LOCATION DOES NOT REPLACE GENETIC CLASSIFICATION

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone	AJANTA / ELLORA → human rock-cut architecture	MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone
BORRA / BELUM → natural caves in Andhra Pradesh	RULE → location does not replace genetic classification	BORRA / BELUM → natural caves in Andhra Pradesh

CORE CLAIM

- MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone
- BORRA / BELUM → natural caves in Andhra Pradesh

MECHANISM

- AJANTA / ELLORA → human rock-cut architecture
- RULE → location does not replace genetic classification

CONSEQUENCE / LIMIT

- MAWMLUH / MAWSMAI / SIJU → Meghalaya limestone
- BORRA / BELUM → natural caves in Andhra Pradesh

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → location does not replace genetic classification.

08

STAGE 08

PROXY PIPELINE

PROXY PIPELINE · RAINFALL, VEGETATION AND SOIL CO2 · ROUTE AND RESIDENCE TIME · ELEMENT SIGNAL · DATED SERIES WITH UNCERTAINTY

CLIMATE → rainfall, vegetation and soil CO2 → RECHARGE → route and residence time → CALCITE → isotope / element signal → CHRONOLOGY → dated series with uncertainty

CORE CLAIM

- CLIMATE → rainfall, vegetation and soil CO2
- RECHARGE → route and residence time

MECHANISM

- CALCITE → isotope / element signal
- CHRONOLOGY → dated series with uncertainty

CONSEQUENCE / LIMIT

- CLIMATE → rainfall, vegetation and soil CO2
- RECHARGE → route and residence time

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CALCITE → isotope / element signal; CHRONOLOGY → dated series with uncertainty.

09

STAGE 09

MEGHALAYAN HIERARCHY

MEGHALAYAN HIERARCHY · KM-A SPELEOTHEM AT MAWMLUH · CLOSE DISTINCTION · SOLUTION ENLARGES VOIDS WHEREAS DEPOSITION BUILDS SPELEOTHEMS; STALACTITES HANG · MEGHALAYAN CHRONOSTRATIGRAPHIC UNIT ARE CATEGORICALLY DIFFERENT.

QUATERNARY → Period

HOLOCENE → Epoch → MEGHALAYAN → Age / Stage → GSSP → KM-A speleothem at Mawmluh

CORE CLAIM

- QUATERNARY → Period
- HOLOCENE → Epoch
- MEGHALAYAN → Age / Stage

MECHANISM

- GSSP → KM-A speleothem at Mawmluh
- CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems
- stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the

CONSEQUENCE / LIMIT

- Meghalayan chronostratigraphic unit are categorically different.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems; stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the.

09

STAGE 09 MEGHALAYAN HIERARCHY

MEGHALAYAN HIERARCHY · KM-A SPELEOTHEM AT MAWMLUH · CLOSE DISTINCTION · SOLUTION ENLARGES VOIDS WHEREAS DEPOSITION BUILDS SPELEOTHEMS; STALACTITES HANG · MEGHALAYAN CHRONOSTRATIGRAPHIC UNIT ARE CATEGORICALLY DIFFERENT.

QUATERNARY → Period

HOLOCENE → Epoch → MEGHALAYAN → Age / Stage → GSSP → KM-A speleothem at Mawmluh

CORE CLAIM
- QUATERNARY → Period
- HOLOCENE → Epoch
- MEGHALAYAN → Age / Stage

MECHANISM
- GSSP → KM-A speleothem at Mawmluh
- CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems
- stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the

CONSEQUENCE / LIMIT
- Meghalayan chronostratigraphic unit are categorically different.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE DISTINCTION: Solution enlarges voids whereas deposition builds speleothems; stalactites hang, stalagmites rise, columns join them, and Meghalaya caves, Bhimbetka shelters and the.

10

STAGE 10 STATUS AND NARRATIVE CAUTION

STATUS AND NARRATIVE CAUTION · 4.2 KA EVENT · CORRELATION GUIDE · GSSP POINT · PHYSICAL LOWER BOUNDARY · NAMED FORMAL INTERVAL · NOT A RATIFIED EPOCH · EVIDENCE LIMIT

CORE CLAIM	MECHANISM	CONSEQUENCE / LIMIT
4.2 ka EVENT → correlation guide	MEGHALAYAN → named formal interval	EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary must not be collapsed
GSSP POINT → physical lower boundary	ANTHROPOCENE → not a ratified Epoch	tourism and infrastructure effects depend on recharge, passage geometry, biodiversity and carrying capacity.

CORE CLAIM
- 4.2 ka EVENT → correlation guide
- GSSP POINT → physical lower boundary

MECHANISM
- MEGHALAYAN → named formal interval
- ANTHROPOCENE → not a ratified Epoch

CONSEQUENCE / LIMIT
- EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary must not be collapsed
- tourism and infrastructure effects depend on recharge, passage geometry, biodiversity and carrying capacity.

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EVIDENCE LIMIT: Cave age, host-rock age, occupation age and formal geological-age boundary must not be collapsed; tourism and infrastructure effects depend on recharge, passage geometry, biodiversity and carrying capacity.

11

STAGE 11 KARST CONSERVATION SPINE

KARST CONSERVATION SPINE · RECHARGE, PASSAGES, SPRINGS AND VALUES · IRREVERSIBLE QUARRY AND POLLUTION PATHWAYS · VISITATION, WASTE AND WATER USE · HYDROLOGY, BIODIVERSITY AND STRUCTURAL CHANGE

MAP → recharge, passages, springs and values → AVOID → irreversible quarry and pollution pathways → MANAGE → visitation, waste and water use → MONITOR → hydrology, biodiversity and structural change

CORE CLAIM
- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

MECHANISM
- MANAGE → visitation, waste and water use
- MONITOR → hydrology, biodiversity and structural change

CONSEQUENCE / LIMIT
- MAP → recharge, passages, springs and values
- AVOID → irreversible quarry and pollution pathways

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: MANAGE → visitation, waste and water use; MONITOR → hydrology, biodiversity and structural change.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT · MEGHALAYA HAS INDIA'S PREMIER KARST SYSTEMS. KREM LIAT PRAH · THE CHERRAPUNJI LIMESTONE CAVE SHOWS WELL-DEVELOPED STALACTITE AND STALAGMITE · CAVE TOURISM SPANS TWO DIFFERENT CATEGORIES · NATURAL KARST CAVES IN MEGHALAYA AND ROCK-CUT/ · VERY HIGH RAINFALL OVER THICK LIMESTONE MAKES MEGHALAYA IDEAL · THESE CAVES DOUBLE AS PALAEOCLIMATE ARCHIVES — THEIR STALAGMITES · GLOBAL BOUNDARY STRATOTYPE SECTION AND POINT — THE PHYSICAL

OPTIONAL SOURCE DEPTH
- Meghalaya has India's premier karst systems. Krem Liat Prah is widely cited as India's longest
- The Cherrapunji limestone cave shows well-developed stalactite and stalagmite formations (dripstone deposition).
- Cave tourism spans two different categories: natural karst caves in Meghalaya and rock-cut/
- Very high rainfall over thick limestone makes Meghalaya ideal for rapid solution and cave growth.

ADVANCED DEBATE
- These caves double as palaeoclimate archives — their stalagmites record past rainfall/drought.
- GSSP = Global Boundary Stratotype Section and Point — the physical 'golden spike' defining a time boundary.
- The Meghalayan Age is the youngest stage of the Holocene (last ~4,200 years); its GSSP is the KNI-51 stalagmite in Mawmluh Cave, Meghalaya.
- The stalagmite isotope signal marks abrupt weakening of the Indian summer monsoon around 4.2 ka.

LEGACY / NUANCE
- Krem Liat Prah in Meghalaya is widely cited as India's longest natural cave; mapped length is survey-dependent.
- Cherrapunji limestone cave: developed stalactite/stalagmite (Khullar).
- Ajanta (Maharashtra), Pachmarhi (MP ~500 caves), Khandagiri/Udayagiri (Odisha).
- High rainfall + thick limestone = rapid karst solution in Meghalaya.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels